AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies



TM341
Computer Vision

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 21/April/2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 120 Minutes**

Instructions:

- Total Marks: 30
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part – A (5 Marks)

1. The input to a computer vision system is usually:
   A) Text
   B) Audio
   C) Images or video
   D) Numbers
2. "There's an object and a background!" refers to:
   A) Low-Level Vision
   B) Mid-Level Vision
   C) High-Level Vision
   D) Image Enhancement
3. Image filtering changes the:
   A) Domain of the image
   B) Range of the image
   C) Coordinates only
   D) Image size only
4. Median filtering is especially effective for removing:
   A) Gaussian noise
   B) Salt and pepper noise
   C) Motion blur
   D) Low contrast
5. Dilation:
   A) Shrinks objects
   B) Detects corners
   C) Expands white regions
   D) Converts to binary

Part – B (10 Marks)

Question 1: (3 marks).

a) Define convolution and kernel.

b) Complete the code to sharpen the image using a custom kernel.

```
import cv2
import numpy as np

img = cv2.imread('image.jpg')

kernel = np.array([[0, -1, 0],
                   [-1, ____, -1],
                   [0, -1, 0]])

sharp = cv2.____________(img, -1, kernel)
```

Question 2: (4 marks)

a) What is the purpose of a sharpening (high-pass) filter in image processing?

b) You are given the following sharpening kernel:

[[0, -1, 0],
 [-1, 5, -1],
 [0, -1, 0]]

Calculate the new value of the center pixel after applying the Gaussian filter.

Assume the 3×3 region around the pixel is:

[[100, 100, 100],
 [100, 120, 100],
 [100, 100, 100]]

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Question 3: (3 marks)

Complete the code for image resizing:
img = cv2.imread('image.jpg')

a) Get original dimensions
h, w = _____

b) Resize image to half its size
resized = cv2.resize(img, (_____, _____))

c) Display result
_____
cv2.waitKey(0)
cv2.destroyAllWindows()

Part – C (15 Marks)

Question 1: (5 marks)

a) What is image stitching in image processing?
b) Explain the basic steps involved in stitching two images together.

Question 2: (5 marks)

An image of size 256×256 is processed in the frequency domain.

a) Write the steps to apply frequency filtering.
b) If a circular mask of radius 20 pixels is removed from the center of the spectrum, what type of filter is applied? Why?
c) What happens visually to the image?

Question 3: (5 marks)

Explain the purpose of the Laplacian operator in image processing. How does it detect edges in a grayscale image, and what is the main difference between the Laplacian and the Sobel operator?

End of questions

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